

PEACH ???

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Abstract—???

Index Terms—???

I. INTRODUCTION

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BC: todo o texto abaixo vc precisa modificar para não termos problemas com plágio do artigo que foi submetido, reescreva a versão nova usando o seu comando de cor "pr"

II. SYSTEM MODEL

The user is located in the xy -plane, considerando a elevação do array é expressa como:

$$\mathbf{u} = [x_u \ y_u \ z_u]^T, \quad \mathbf{p} = [x_m \ y_m \ (z_m + z_{elev})]^T, \quad (1)$$

Se vc considera o usuario no plano XY faz sentido colocar z_u ? where $\mathbf{u}$ represents the location of the user, and $\mathbf{p}_m$ represents the location of the m -th antenna.

The reference antenna position at the bottom-left corner a_{11} is $(0, 0, z_{elev})$. faça um plot das posições do array para averiguarmos essa afirmação!

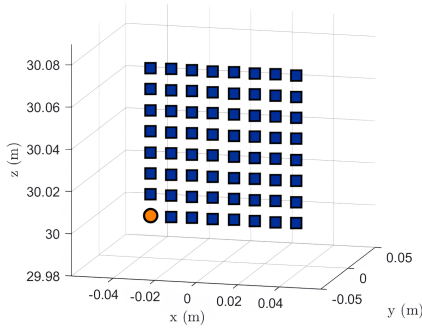


Figure 1: Caption

Where, given the pre-established conditions in this presentation, we have:

$$a_{mn} = e^{-j \frac{2\pi}{\lambda} \sqrt{[x_u - (m \cdot \delta_x)]^2 + y_u^2 + [z_u - (n \cdot \delta_z + z_{elev})]^2}} \quad (2)$$

faltou vc definir λ , δ_x e δ_z

Texto abaixo misturando ingles com portugues !

O path loss pode ser fragmentado em três termos, path loss, shadowing e fading. Dessa forma cada termo possui

diferentes variações na amplitude do sinal, devido a área em que é coletada a média de cada termo. The area mean, é o termo longo do path loss, caracterizando uma variação lenta, e mede a atenuação média da amplitude do sinal em relação à distância, sendo esse determinístico. Esse paper assume um cenário Line-of-Sight (LoS) puro, portanto o termo longo é o suficiente para representar os efeitos do path loss no modelo de canal. É necessário adotar altas frequências para viabilizar o uso de sistemas M-MIMO a partir da miniaturização das antenas, portanto, deve ser incluído large-scale fading coefficient (g).

por favor, clarifique: 1.) Por que, especificamente, alta frequência viabiliza M-MIMO? 2.) O que 'termo longo' representa no seu contexto (apenas path loss?) e como se alinha com LoS puro e a inclusão de g ? 3.) Como a alta frequência influencia seu modelo de LSF (ou a eq. 3)?

$$g_{ij} = \Upsilon \left(\frac{1}{d_{ij}} \right)^\alpha \quad (3)$$

where $\Upsilon = \left(\frac{\lambda}{4\pi} \right)^\alpha$ is the channel gain at a 1 meter reference distance baseado na menor área da antena dado a frequencia do sistema. and (α) is the path loss exponent.[1].

Construimos o processo aleatório $\mathbf{Y} \in \mathbb{C}^{M \times N}$ com as séries temporais dos sinais recebidos em cada antena;

$$\mathbf{Y} = e^{\psi_0} \sqrt{\mathbf{G}} \mathbf{A} \cdot \mathbf{S} + \mathbf{N}, \quad \mathbf{N} \sim \mathcal{N}_{\mathbb{C}}(0, \sigma^2 \mathbf{I}_M) \quad (4)$$

Portanto $\mathbf{Y}$ é caracterizado a partir de suas grandes médias. Modelando um usuário estático e Additive White Gaussian Noise (AWGN), $\mathbf{Y}$ é processo ergódico, e pelo menos Wide-Sense Stationarity - WSS.

Ficou faltando a definição formal de $\mathbf{S}$,

Ficou faltando fazer o link entre os elemento de $\mathbf{g}$ e a matriz $\sqrt{\mathbf{G}}$, pra vc como essa raiz quadrada é calculada ?

Quais propriedades de $\mathbf{S}$ e dos coeficientes de canal ($\mathbf{A}$, $\mathbf{G}$) são necessárias para que $\mathbf{Y}$ seja WSS?

Revise sua interpretação de Υ , "menor area de uma antena" ? O que você quer dizer com ' $\mathbf{Y}$ é caracterizado a partir de suas grandes médias'? Poderia clarificar essa expressão?

In this study, we investigate the precise localization of a single user within a defined area using an elevated antenna array. As illustrated in Fig. ?? faça uma figura sua aqui pra representar o modelo de sistema, the antennas are positioned atop a building, mimicking a realistic urban base station deployment. This elevated configuration not only improves

spatial diversity, by providing a richer angular view compared to scenarios where antennas and users share the same height, but also allows the conventional Multiple Signal Classification (MUSIC) algorithm to estimate full three-dimensional coordinates. In contrast, traditional setups with co-located heights typically yield only the Direction of Arrival (DoA), necessitating modifications to infer precise vertical positioning. Thus, the elevated array enables more accurate estimation of the user's horizontal (x, y) and vertical (z) coordinates, enhancing overall localization performance.

In this analysis, we consider an antenna array confined to the xz -plane, with the user's location confined to the ground-level xy -plane. Hence, the positions of the user and each antenna element are represented, respectively, by:

$$\mathbf{u} = [x \ y \ 0]^T, \quad \mathbf{p}_n = [x_n \ 0 \ z_n]^T, \quad n = 1, \dots, N, \quad (5)$$

where $\mathbf{u}$ indicates the user's position, and $\mathbf{p}_n$ specifies the location of the n -th antenna element. For elevated arrays such as the Uniform Planar Array (UPA), we define $\bar{z}_n$ as the average height of the antenna configuration.

Given this configuration, our goal is to accurately estimate the user's location from the observed signal vector $\mathbf{r}(t)$, which is inherently linked to the steering vector $\mathbf{a}(\mathbf{u})$. The steering vector depends on the Euclidean distances between the antenna elements and the user location, alongside path loss characteristics of the propagation channel. We consider the following radar-based signal model:

$$\mathbf{r}(t) = e^{j\psi_0} \mathbf{a}(\mathbf{u}) s(t) + \mathbf{n}(t), \quad (6)$$

where $\mathbf{r}(t)$ denotes the received signal vector at the antenna elements, $s(t)$ represents the transmitted signal from the user, and $\mathbf{n}(t)$ is the additive noise vector, with each component $n_i(t)$ assumed to be independent and identically distributed (i.i.d.) complex Gaussian noise with variance σ_n^2 . For accurate modeling of Line-of-sight (LOS) conditions at higher frequencies, our channel model incorporates the free-space path loss factor $\lambda/(4\pi)$, ensuring practical relevance and analytical clarity as validated by recent Integrated Sensing and Communication (ISAC) channel characterization studies [?], [?], [?]. The phase offset ψ_0 indicates the differential phase of the reflected signal relative to a reference antenna element, typically the first antenna in the array.

The steering vector $\mathbf{a}(\mathbf{u})$, capturing both directional information and path-loss attenuation, is explicitly expressed as:

$$\mathbf{a}(\mathbf{u}) = \frac{\lambda}{4\pi} \left[\frac{1}{d_1}, \frac{e^{-j\frac{2\pi}{\lambda}(d_2-d_1)}}{d_2}, \dots, \frac{e^{-j\frac{2\pi}{\lambda}(d_N-d_1)}}{d_N} \right]^T, \quad (7)$$

where d_i represents the Euclidean distance between the user and the i -th antenna element.

We assume a pure LOS scenario, appropriate for elevated antenna deployments typical in urban ISAC scenarios, especially at higher frequencies such as 15 GHz, where the LOS

component typically dominates due to higher path loss in Non-Line-of-sight (NLOS) paths

III. CRAMÉR-RAO BOUND (CRB) FOR LOCALIZATION

To rigorously evaluate the theoretical accuracy limits of the proposed localization approach, we derive the Cramér-Rao Bound (CRB) for estimating the user's two-dimensional position $\mathbf{u} = [x, y]^T$. The CRB is computed using the Fisher Information Matrix (FIM), quantifying the amount of information contained in the received signals relative to the position parameters.

Considering phase measurements at each antenna element, the CRB for coordinates x and y is expressed as:

$$\text{CRB}(x) = \left[J_{xx} - \frac{J_{xy}^2}{J_{yy}} \right]^{-1}, \quad \text{CRB}(y) = \left[J_{yy} - \frac{J_{xy}^2}{J_{xx}} \right]^{-1}, \quad (8)$$

with FIM elements defined as:

$$J_{xx} = P_{\text{norm}} \sum_{n=1}^N \left[\frac{\delta_n^2}{d_n^6} + \frac{4\pi^2}{\lambda^2} \left(\frac{\delta_n}{d_n^2} - \frac{\delta_1}{d_1 d_n} \right)^2 \right], \quad (9)$$

$$J_{yy} = P_{\text{norm}} y^2 \sum_{n=1}^N \left[\frac{1}{d_n^6} + \frac{4\pi^2}{\lambda^2} \left(\frac{1}{d_n^2} - \frac{1}{d_1 d_n} \right)^2 \right], \quad (10)$$

$$J_{xy} = P_{\text{norm}} y \sum_{n=1}^N \left[\frac{\delta_n}{d_n^6} + \frac{4\pi^2}{\lambda^2} \left(\left(\frac{\delta_n}{d_n^2} - \frac{\delta_1}{d_1 d_n} \right) \times \left(\frac{1}{d_n^2} - \frac{1}{d_1 d_n} \right) \right) \right], \quad (11)$$

where $\delta_n = x - x_n$, $\delta_1 = x - x_1$, and $d_n = \sqrt{(x - x_n)^2 + y^2 + z_n^2}$ represents the Euclidean distance between the user and each antenna element. The normalized power factor is defined as $P_{\text{norm}} = L \left(\frac{|s|\lambda}{\sqrt{2\pi\sigma_n}} \right)^2$, with L being the number of samples, $|s|$ the transmitted signal amplitude, and σ_n^2 the noise variance.

Notably, due to the dependence on antenna geometry and elevation, the CRB for coordinates x and y exhibits asymmetry, significantly impacting the localization accuracy according to antenna elevation and user positioning, as numerically analyzed in Section VII.

IV. APPLICATION OF THE MUSIC ALGORITHM IN MULTIPLE ANTENNA SYSTEMS

In MIMO systems each antenna in the system architecture contributes to signal reception, combining the signals received by the various antennas in the array. In terms of localization, it is possible to analyze, express and represent directions in the space from linear combinations of the antennas. In AoA estimation algorithms such as MUSIC and ESPRIT, the directions of interesting are only those with user signal components **MUSIC utiliza o subespaço de ruído pra estimar e não o do sinal.** In the MUSIC, the AoA estimation is done by separating the noise subspace ($\mathbf{U}_N$) and the signal subspace ($\mathbf{U}_S$) **Como é feita essa separação ?** . The signals incident **(o que vc entende como " signals incident " ?)** on the antenna array are estimated though the covariance matrix ($\mathbf{C}_Y$) of the

received signal matrix ($\mathbf{Y}$), which contains the covariances between each pair of elements in a set of variables. This allows for the analysis of the correlation between the received signals. $\mathbf{C}_Y$ is defined by eq. (12)

$$\begin{aligned}\mathbf{C}_Y &= \mathbb{E}[(\mathbf{Y} - \mu_Y)(\mathbf{Y} - \mu_Y)^H] \\ &= \mathbb{E}[\mathbf{Y}\mathbf{Y}^H] = \mathbf{\Gamma}_Y \text{Suposição Implícita: Qual suposição sobre a média do sinal recebida permite esta simplificação?} \\ &= \mathbb{E}[(\mathbf{G}\mathbf{A}\mathbf{S} + \mathbf{N})(\mathbf{G}\mathbf{A}\mathbf{S} + \mathbf{N})^H] \\ &= \mathbb{E}[\mathbf{G}\mathbf{A}\mathbf{S}\mathbf{S}^H\mathbf{A}^H\mathbf{G}^H] + \mathbb{E}[\mathbf{G}\mathbf{A}\mathbf{S}\mathbf{N}^H] \\ &\quad + \mathbb{E}[\mathbf{N}\mathbf{S}^H\mathbf{A}^H\mathbf{G}^H] + \mathbb{E}[\mathbf{N}\mathbf{N}^H] \\ &= \mathbb{E}[\mathbf{G}\mathbf{A}\mathbf{S}\mathbf{S}^H\mathbf{A}^H\mathbf{G}^H] + \mathbb{E}[\mathbf{N}\mathbf{N}^H] \text{Justificativa da Simplificação: Este passo elimina os termos cruzados. Qual suposição fundamental?} \\ &= \mathbf{G}\mathbf{A}\mathbf{\Gamma}_S\mathbf{A}^H\mathbf{G}^H + \mathbf{\Gamma}_N \text{Definição das Covariâncias: Aqui surgem } \mathbf{\Gamma}_S \text{ e } \mathbf{\Gamma}_N. \text{ Como elas são formalmente definidas a partir dos termos?} \\ &= \mathbf{G}\mathbf{A}\mathbf{\Gamma}_S\mathbf{A}^H\mathbf{G}^H + \sigma^2\mathbf{I} \text{Modelo do Ruído: A substituição de } \mathbf{\Gamma}_N \text{ por } \sigma^2\mathbf{I} \text{ é um modelo específico. Quais propriedades do ruído } \mathbf{N} \text{ s}\end{aligned} \quad (12)$$

REVISADO ATÉ AQUI!

where $\mathbf{G}^{\frac{1}{2}} = \text{diag}(\sqrt{g_1}, \sqrt{g_2}, \dots, \sqrt{g_m})$. $\mathbf{A}$ is the steering vector, $\mathbf{X}$ is the matrix of transmitted signals, and $\mathbf{N}$ is the noise matrix. $\mathbf{C}_Y$ descrição da relação estatística entre os sinais recebidos nos M elementos. Through $\mathbf{C}_Y$, the eigendecomposition of a matrix is performed, as shown in eq. (13), where the matrix is represented in terms of its eigenvalues and eigenvectors.

$$\mathbf{R}_Y = \mathbf{U}\mathbf{\Sigma}\mathbf{U}^H = \mathbf{U}_s\mathbf{\Sigma}_s\mathbf{U}_s^H + \mathbf{U}_n\sigma^2\mathbf{U}_n^H \quad (13)$$

where $\mathbf{U}$ is a vector space, which can be understood as an orthonormal matrix since each column represents an eigenvector orthogonal to the other eigenvectors of the field [2]. Aplicando a decomposição espectral nos temos;

$$[\mathbf{U}, \mathbf{\Sigma}] = \text{eig}(\mathbf{R}_Y) \quad (14)$$

The number of antennas that an array contains within its envelope defines the maximum DoF (Degrees of Freedom) of a system, equivalent to the same number of eigenvectors in $\mathbf{U}$, i.e., $v_1, v_2, \dots, v_M \in \mathbf{U}$. Thus, each eigenvector defines a decorrelated spatial direction. $\mathbf{\Sigma}$ is the diagonal matrix that contains the eigenvalues corresponding to the eigenvectors of $\mathbf{U}$. The eigenvectors have power associated with each component, allowing one to determine which directions have greater contributions.

Let $\mathbf{U}_s$ and $\mathbf{U}_n$ be vector subspaces defined over the field $\mathbf{U}$, which is also a vector space. $\mathbf{U}_n$ is formed by the eigenvectors corresponding to the small eigenvalues, and these eigenvectors correspond to the noise components.

$$P_{\text{MUSIC}}(\mathbf{u}_i) = [\mathbf{a}(\mathbf{u}_i)^H \hat{\mathbf{U}}_n \hat{\mathbf{U}}_n^H \mathbf{a}(\mathbf{u}_i)]^{-1} \quad (15)$$

A. Increasing spatial diversity with elevated Array

B. Effect of Elevation on the MUSIC Pseudo-Spectrum

Elevation differences between antenna arrays and users profoundly impact the shape and localization characteristics of the MUSIC pseudo-spectrum. Unlike traditional configurations,

where arrays and users are co-planar, elevated antenna arrays exhibit distinct pseudo-spectral patterns that significantly enhance localization capability and accuracy. To illustrate this, we consider a numerical scenario involving a UPA consisting of $N = 64$ antennas arranged in an 8×8 grid. Operating at a carrier frequency of 15 GHz, we evaluate the MUSIC pseudo-spectrum for a user located at 30 meters from the origin in the xy plane with an Signal-to-Noise Ratio (SNR) of 15 dB and minimal sampling ($L = 2$).

V. POSITION ESTIMATION USING PEACH-MUSIC

VI. POSITION ESTIMATION ALGORITHM USING CIRCLES

AND HYPERBOLAS IN PEACH-MUSIC APPROACH
The MUSIC pseudo-spectrum from a horizontal Uniform Linear Array (ULA) (antennas along the x -axis) produces a hyperbolic detection region, while a vertical ULA (antennas along the z -axis) yields a circular region in the xy -plane. The intersection of these regions, illustrated in Fig. ??, accurately determines the user's position. Considering a Uniform Rectangular Array (URA) with $N_x \times N_y$ elements as two orthogonal ULAs, Position Estimation Algorithm using Circle and Hyperbola in MUSIC (PEACH-MUSIC) exploits this geometry for efficient localization. The vertical array provides a circle:

$$x^2 + y^2 = d_{xy}^2, \quad (16)$$

with radius d_{xy} directly estimated from the vertical ULA pseudo-spectrum.

The horizontal array defines a hyperbola based on distance differences (Δ) to antenna endpoints at symmetric positions $(-c, 0, z_a)$ and $(c, 0, z_a)$:

$$|d_1 - d_2| = \Delta, \quad (17)$$

with

$$d_{1,2} = \sqrt{(x \mp c)^2 + y^2 + z_a^2}. \quad (18)$$

Solving analytically, the user's coordinates are $(x, y, 0)$, where:

$$x^2 = \frac{\Delta^2}{16c^2} [4(c^2 + d_{xy}^2 + z_a^2) - \Delta^2], \quad (19)$$

and

$$y^2 = d_{xy}^2 - x^2. \quad (20)$$

Coordinate signs are defined by MUSIC pseudo-spectrum peaks, significantly reducing computational effort compared to grid searches.

A. Efficient User Position Estimation

We propose a simplified one-dimensional scan for efficient localization. Fixing $y = 0$, the vertical and horizontal ULA spectra are sampled at N_{circle} and $N_{\text{hyper}} = 2N_{\text{circle}}$ points, respectively. Identifying the spectral peaks, the user's position is computed directly from the closed-form equations above, resolving coordinate signs using the horizontal ULA spectrum.

Algorithm 1 summarizes this efficient localization process, emphasizing its computational simplicity and practical utility.

Algorithm 1 PEACH-MUSIC Algorithm

- 1: **Input:** Received signals from $N_x \times N_y$ URA, $x_{\max}$, N_{circle} , $N_{\text{hyper}} = 2N_{\text{circle}}$
- 2: **Output:** Estimated position $(\hat{x}, \hat{y})$
- 3: Estimate Δ from horizontal ULA pseudo spectrum line ($y = 0$)
- 4: Estimate d_{xy} from vertical ULA pseudo spectrum line ($y = 0$)
- 5: Compute user's coordinates $(x, y, 0)$ analytically:

$$x = \pm \sqrt{\frac{\Delta^2}{16c^2} [4(c^2 + d_{xy}^2 + z_a^2) - \Delta^2]}; \quad y = \sqrt{d_{xy}^2 - x^2}$$

- 6: Resolve the sign of x using horizontal ULA pseudo spectrum
 - 7: **return** $(\hat{x}, \hat{y})$
-

This estimation strategy significantly reduces computational complexity while ensuring precise and efficient localization.

VII. NUMERICAL RESULTS

TABLE I
SYSTEM AND PARAMETERS SIMULATION

Parameters	Values
Realizations	$\mathcal{M} = 500$
Snapshots	$N = 10^5$
User Position	$\mathbf{u} = [10, 30, 0]$ m
System Architecture	8×8 URA, $\delta = 0, 5\lambda$
Carrier Frequency	$f_c = 15.0$ Ghz
Transmitted signal power	$\mathcal{P} = 100$ mW
Path-loss Expoent	$\alpha = 2$
SNR	SNR = 10 dB
Array Elevation	$z = [1 : 10 : 100]$ meters

Coloque aqui os resultados obtidos até o momento, É importante identificar na legenda ou em uma tabela, todos os parametros utilizados !

VIII. CONCLUSION

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REFERENCES

- [1] T. L. Marzetta, *Fundamentals of Massive MIMO*, 1st ed. Cambridge, UK: Cambridge University Press, 2016.
- [2] E. Björnson, Ö. T. Demir *et al.*, "Introduction to multiple antenna communications and reconfigurable surfaces," 2024.